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DRUG METABOLISM

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Overview

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Introduction

A knowledge of the fate of a drug, its disposition (absorption, distribution, metabolism, and excretion; ADME) and pharmacokinetics (PK) (the mathematical description of these processes) is important throughout pharmaceutical research and development. The aim of a typical pharmaceutical company is to identify and develop good orally active candidate molecules that have suitable potency and selectivity against a target, sufficient oral bioavailability, metabolic stability, and duration of action. The importance of ADME/PK has been highlighted in surveys that have revealed that the main causes of failure during development are unacceptable clinical efficacy, toxicity, and poor PK. In fact, as much as 40% of all failures have been attributed to poor PK. *In silico* (computer-based), *in vitro* (cell-based), and *in vivo* (animal-based) techniques are

available for investigating the ADME/PK of new chemical entities (NCEs). Both the scientific concepts underpinning this discipline and the techniques and rationale employed will be detailed in this article.

The Science of ADME/PK

All drugs have the potential to produce unwanted effects. In drug research and development it is essential to select chemicals that have a margin of safety between the dose that produces the desired (therapeutic) effects and the dose that produces undesired (toxic) effects.

The investigation of ADME/PK aids in the discovery and selection of new drugs, supports safety assessment, and helps define conditions for effective use in patients. The main objectives of ADME/PK studies can therefore be summarized as shown in Table 1.

Drug Absorption

For a drug to work it must be a solution at the site of delivery. If orally administered it must dissolve in the lumen of the small intestine. Once in solution it must

Table 1 Main objectives of ADME/PK studies

To characterize the basic pharmacokinetic parameters of a test substance
To understand the mechanism of action of a drug
To support toxicology studies
To validate the appropriateness of the animal species used in toxicology
To establish margins of safety between the therapeutic and toxic effects
To optimize dosage regime (size, formulation, frequency)
To establish target organs/modes of absorption and elimination
To establish nature of metabolism
To provide data for regulatory authorities

be absorbed through the walls of the intestine and subsequently cross the barriers between the blood stream and the site of action. Absorption can therefore be defined as the net transfer of a substance from its site of administration to the site of effect.

The entry of many orally administered drugs into the systemic circulation is dependent on their tendency to partition into lipid (lipophilicity). Lipophilic properties facilitate a compound's penetration of lipid membranes and transportation in blood lipoproteins but their lower water solubility reduces urinary excretion. In addition, lipophilic compounds are more likely to undergo extensive metabolism.

The degree of ionization of a drug can dictate how lipid soluble it is. Ionization, as measured by the dissociation constant (pK_a), can vary with the pH of the environment. When the pH of a solution is equal with pK_a the concentration of the ionized species is the same as that of the unionized species. As many drugs are either weak acids or weak bases, the degree of ionization can impact on their absorption, distribution, and excretion characteristics as shown in Figure 1.

The solubility of a compound is a barrier that limits its concentration in a solution and therefore the amount of material available for absorption. A compound's crystal type (polymorph), its particle size, wettability, and the use of solubilizing agents in formulations can all influence solubility. Also, the formation of salts, esters, and complexes with a compound can improve its solubility. Solubility is a key parameter for determining the dissolution rate of compounds, a more informative measure that takes into account the form and particle size distribution in a particular medium.

A number of mechanisms of absorption exist but the most important of these, for the vast majority of compounds, is passive diffusion where the movement of compound is driven by the concentration gradient that can occur between cell junctions (paracellular)

or through cells (transcellular). Lipophilic compounds tend to cross the plasma membrane most readily and utilize the transcellular route, while hydrophilic compounds tend to utilize the paracellular route. In addition, some substances can be transported by an active system either with or against a diffusion gradient. An important transporter protein, P-glycoprotein (P-gp), is present in the plasma membrane of many cells and is responsible for the efflux of a wide range of substrates. The presence of a significant P-gp efflux capacity in the intestinal lumen reduces the overall absorption of many compounds. The main mechanisms of transport that can occur in the intestinal epithelium are shown schematically in Figure 2.

The small intestine is folded into villi and microvilli which increases surface area many times. Even poorly soluble compounds often undergo reasonable absorption because of the large surface area. Absorption can be reduced if blood flow in the gastrointestinal tract is insufficient to maintain 'sink conditions' or if transit times within the tract are too rapid.

Drug Bioavailability

Following absorption, first-pass metabolism can reduce the total exposure of the body to drug. First-pass metabolism refers to any loss of the administered material by transmucosal or hepatic means after absorption and before reaching the systemic circulation, and this is shown schematically in Figure 3.

Bioavailability is a term used to describe the systemic availability of drug and can be defined as the rate and extent of appearance of unchanged drug in the systemic circulation following an extravascular (e.g., oral) dose. Experimentally, it is determined as the fraction of the maximal levels of drug present in the systemic circulation after an intravenous dose. It takes into account both absorption and metabolism and is dependent on area under the curve (AUC), peak concentration achieved (C_{max}), and time to reach peak concentration (t_{max}). It is generally expressed as a percentage and is represented by eqn [1]:

$$\% \text{Bioavailability} = \frac{\text{AUC}_1 \times \text{Dose}_2}{\text{AUC}_2 \times \text{Dose}_1} \times 100 \quad [1]$$

where 1 refers to an oral dose and 2 refers to an intravenous dose.

The assessment of many PK parameters depends upon the accurate measurement of AUC. Although software is used routinely to do this, the calculations are straightforward and are based on the sum of

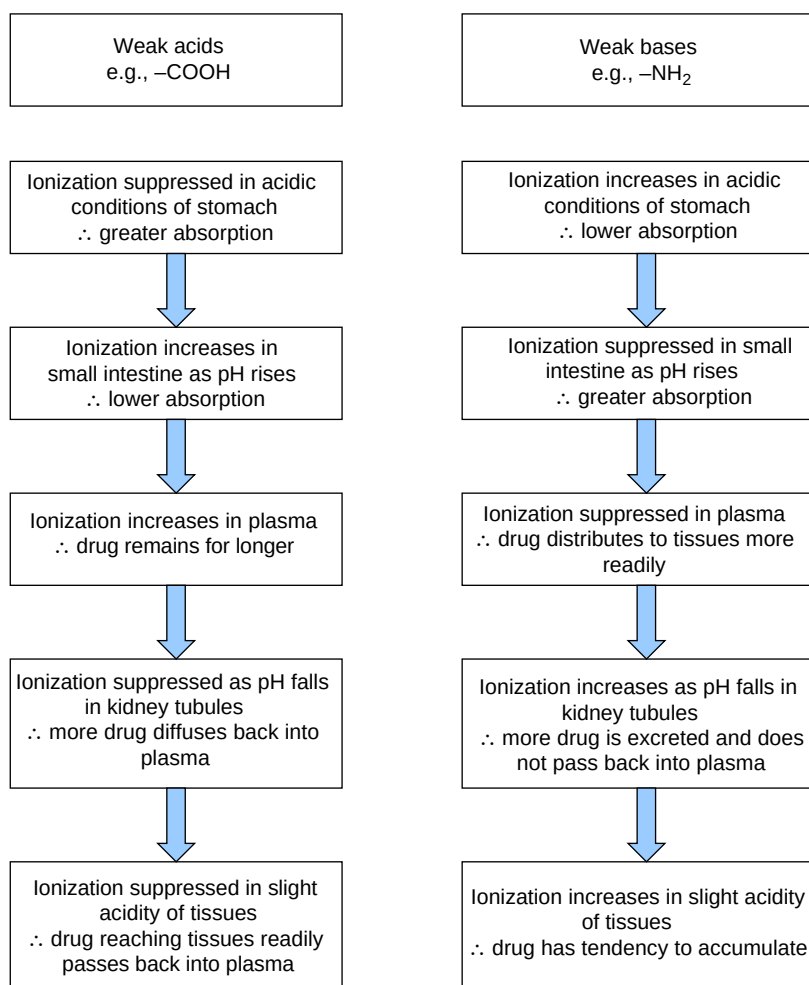


Figure 1 Consequences of ionization for weak acids or bases during their passage through the body.

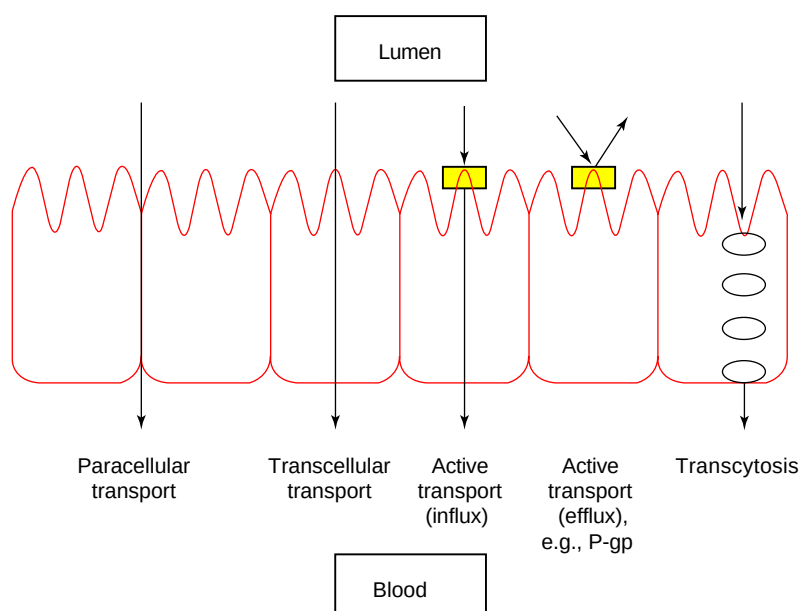


Figure 2 Multiple mechanisms of transport through the intestinal epithelium.

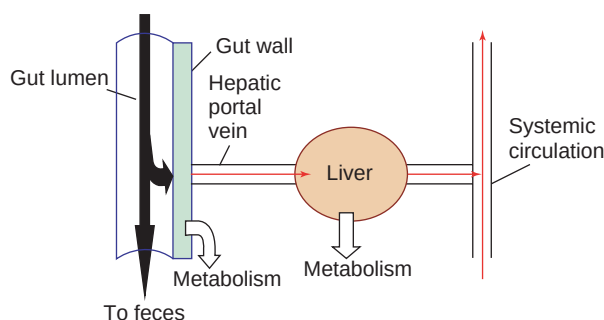


Figure 3 Schematic diagram to show sites of first-pass metabolism.

trapezoid areas under the concentration versus time curve for each sampling interval (trapezoidal rule).

As pharmacological effect depends upon the quantity of unchanged drug reaching the target organ and its residence time, it is evident that changes in absorption and metabolism will lead to changes in bioavailability and therefore activity.

Relative bioavailability (bioequivalence) is the comparison of the rate and extent of drug appearance in the systemic circulation when one formulation is compared with another. When two or more formulations of the same drug produce statistically indistinguishable rates and extents of appearance they can be said to be bioequivalent.

Drug Distribution

Drug distribution can be defined as the (reversible) transfer of drug around the body. This is usually very quick within the vascular system but variable into tissues, being dependent on a number of factors including blood perfusion rate, drug diffusion rate, plasma protein binding, and tissue binding.

The amount of drug distributed cannot normally be measured directly and so is inferred from PK analysis of the plasma concentration–time profile utilizing eqn [2]. The apparent volume of distribution (V_d) can be defined as the volume that a drug would occupy were the drug present in all tissues at the same concentration as observed in the plasma:

$$V_d = \frac{\text{amount in body (dose)}}{\text{plasma concentration } (C_0)} \quad [2]$$

where C_0 is the plasma concentration if drug had been instantaneously distributed into its final volume following an intravenous dose; V_d can vary from 0.05 to 800 l kg⁻¹. The lower values are usually for drugs that remain in the plasma compartment, i.e., they are hydrophilic and/or tightly bound to plasma. The higher values are for drugs that have a high affinity for tissues.

When a membrane is not a substantial barrier to a drug, the rate of perfusion of the tissue becomes the limiting factor; this is commonly the case for small lipophilic molecules. Distribution occurs rapidly in highly perfused organs (e.g., lungs) and slowly in poorly perfused organs (e.g., fat). A diffusion rate limitation occurs when a tissue membrane is a substantial barrier to the movement of drug, e.g., the blood–brain barrier has poorly permeable capillary membranes that are effective in blocking the passage of many drugs into the brain. The rate of distribution is also dependent on total body mass; therefore, distribution equilibrium is likely to be achieved faster in smaller species.

As the most abundant plasma protein is albumin (which prefers to bind acidic drugs), the total binding for acidic drugs can be very high (greater than 99.9%) and therefore the volumes of distribution can be very low. Less abundant but still important is the plasma protein α 1-acid glycoprotein that tends to bind basic drugs. The extent to which a drug binds is governed by an equilibrium (or affinity) constant, the concentration of the protein and of the drug. Therefore, as only unbound ‘free’ drug will pass through membranes, the dissociation rate can control the rate of distribution. In addition to plasma protein binding, drugs can bind to various sites within tissues. Lipophilic weak bases, for instance, have a high affinity for the acidic phospholipids, which make up biological membranes, and therefore they tend to have high volumes of distribution.

Drug distribution is related, as shown in eqn [3], to the plasma elimination half-life ($t_{1/2}$), the time it takes for the plasma concentration or the amount of drug in the body to be reduced by 50%:

$$t_{1/2} = \frac{V_d}{Cl_p} \times 0.693 \quad [3]$$

where Cl_p refers to plasma clearance (see eqn [6]).

The $t_{1/2}$ will therefore increase if V_d increases or Cl_p decreases, and if a drug has a high V_d and a long $t_{1/2}$ this can indicate tissue accumulation.

Drug Elimination

Drug elimination can be defined as the (irreversible) transfer of a drug from the site of measurement (usually plasma or blood) by either excretion (e.g., renal, biliary, pulmonary, sweat and milk excretion) or metabolism.

Renal excretion is particularly important. Glomerular filtration removes unbound low molecular weight compounds and active secretion and passive reabsorption processes occur throughout the kidney tubules depending on a drug’s ability to cross lipid

membranes and degree of ionization. The net rate of renal excretion can be estimated using eqn [4]:

$$\begin{aligned} \text{Net rate of renal excretion} = & \text{Rate of filtration} \\ & + \text{Rate of secretion} \\ & + \text{Rate of reabsorption} \quad [4] \end{aligned}$$

Biliary excretion results in the elimination of drugs in the bile and is determined by molecular weight. Little biliary excretion occurs for molecules with molecular weight less than 300 and very large molecules such as proteins. The approximate molecular weight thresholds above which drugs can be excreted directly in bile are higher in man and monkey (500) than rabbit (475), guinea-pig (400), and rat and dog (325). It is important to note that because the threshold for rat and dog, the usual toxicology species, is somewhat lower than for man, this can lead to species differences for compounds of intermediate molecular weight.

Enterohepatic recycling is where compounds are returned to the gut via the bile and then reabsorbed. This process can lead to very long $t_{1/2}$ values, e.g., $t_{1/2}$ for digoxin in dogs is raised from 6 to 14 h by recycling.

Expired air is an important route of elimination for anesthetics and some volatile compounds are readily excreted in sweat. The elimination of compounds in mother's milk can lead to high exposures for the baby when it is their only source of nourishment. The mucosal secretion of drug back into the gut can also be an important route of elimination.

Clearance is an important property of a drug, and as already indicated in eqn [3], reducing clearance can extend the $t_{1/2}$. It can be defined as the volume of blood (or plasma) from which all drug is removed in unit time and depends upon the flow of blood through organs and their extraction ratio (ER). The ER can be calculated from drug concentrations as shown in eqn [5]:

$$ER = \frac{C_{in} - C_{out}}{C_{in}} \begin{cases} \text{If organ extracts nothing } ER = 0 \\ \text{If organ extracts everything } ER = 1 \end{cases} \quad [5]$$

where C_{in} and C_{out} are the concentrations of drug entering and leaving the organ, respectively. In the case of the liver, C_{in} is the concentration in the hepatic portal vein blood and C_{out} the concentration in the systemic circulation.

The total clearance is the sum of the clearances in all organs. If the clearance of drug from blood/plasma approaches the blood/plasma flow through the liver, this indicates that the drug is likely to undergo hepatic first-pass removal. An estimate of Cl_p

can be obtained from intravenous dosing data using eqn [6]:

$$Cl_p = \frac{\text{DOSE}}{\text{AUC}} \quad [6]$$

During a drug's development it is important to investigate the effect on exposure of changing the dose, and to prove that clearance is dose-independent. When Cl_p , V_2 , and $t_{1/2}$ remain constant over a dose range and over time, a drug can be said to have linear PK. If an elimination process becomes saturated, AUC tends to increase with dose more than one would expect, if absorption or plasma protein binding are saturated, AUC tends not to increase with dose as much as one would predict.

Knowing the Cl_p and $t_{1/2}$ of a drug one can modify the size and frequency of doses. Doses should be given at repeated intervals such that the 'rate in' equals the 'rate out'. If doses are too high and frequent, then toxic levels may be achieved. If doses are too low and infrequent, then the drug may prove ineffective. **Figure 4** shows the ideal situation where plasma concentrations reach a plateau and the same maximum (C_{max}) and minimum (C_{min}) concentrations are reproduced within the therapeutic window, i.e., C_{min} is kept above the minimum effective concentration and C_{max} below the minimum toxic concentration.

While PK data indicate what the body does to the drug over time, pharmacodynamic (PD) data indicate what the drug does to the body over time. PK/PD modeling involves linking PK and PD data to assess the response-time relationship. The predictivity of models varies with their complexity and the type of data available. Often, response does not track directly with concentrations with the greatest effect being observed well after maximal concentrations have occurred. Generally, the most reliable models depend on the measurement of 'free' levels at the site

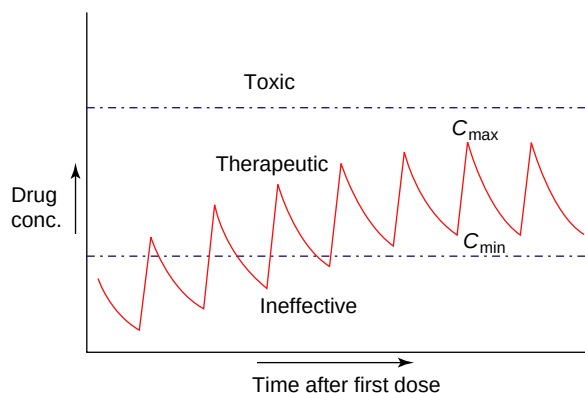


Figure 4 Plot of plasma concentration versus time following the application of an 'ideal' dosing regimen.

of drug action and the accurate and precise measurement of drug effect.

Drug Metabolism

Drugs are metabolized if the body is unable to excrete them unchanged. The main site of metabolism is the liver. The principal reason for metabolism is to increase the water solubility and hence the excreatability of the compound. It is important to note that many of the enzymes involved in drug metabolism have evolved for the metabolism of naturally occurring compounds and only attempt to metabolize drugs because they resemble them in some way. The route of metabolism of a drug can determine whether it shows any pharmacological or toxicological activity. Usually, the metabolism of a drug reduces its toxicity but certain metabolites (usually short-lived, reactive intermediates) are able to produce toxic effects. Studying metabolism allows species comparison of metabolite patterns to validate toxicity studies, to help understand PK/PD, and to allow extrapolation to man. The PK of metabolites along with that of the parent compound may help to give a better overall understanding of the observed effect (desired or undesired).

The main routes of metabolism are the chemical reactions of oxidation, reduction, hydrolysis, and conjugation. These can be divided into two phases: phase I (or functionalization reactions) and phase II (or conjugative reactions).

Phase I metabolism Phase I reactions (mainly oxidation, reduction, and hydrolysis) act as a preparation of the drug for the phase II reactions, i.e., a chemically reactive group is produced or uncovered on which the phase II reactions can occur, e.g., $-\text{OH}$, $-\text{NH}_2$, $-\text{SH}$, $-\text{COOH}$. Most toxic metabolites are produced by phase I reactions. The P-450 isoenzymes (CYP enzymes), known collectively as the mixed function oxidase system, are found in the endoplasmic reticulum of many cells (notably those of liver, kidney, lung, and intestine) and perform many of these different functionalization reactions. The system requires the presence of molecular oxygen and co-factor nicotinamide adenine dinucleotide phosphate (NADPH) as well as cytochrome P450, NADPH-cytochrome P450 reductase, and lipid.

Cytochrome P450 is a heme-containing protein embedded in the endoplasmic reticulum, which exists in multiple forms and has a molecular weight of 45 000–55 000. The heme (Figure 5) is noncovalently bound to the apoprotein and serves as both the oxygen and substrate-binding site.

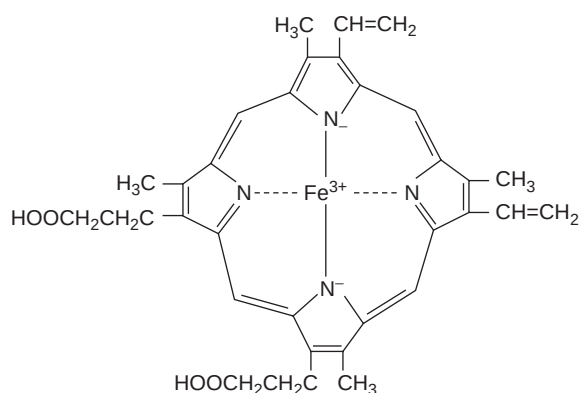


Figure 5 Ferriprotoporphyrin IX heme prosthetic group.

The oxidation reaction catalyzed by CYPs conforms to the stoichiometry shown in eqn [7]:



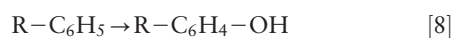
where RH represents an oxidizable drug substrate and ROH is the hydroxylated metabolite. The source of the reducing equivalent (2H^+ , 2e^-) is either NADPH or NADH mediated by a flavoprotein (FAD).

Three CYP families, CYP1, CYP2, and CYP3, account for ~70% of human hepatic microsome CYPs with CYP3 accounting for ~30%. These CYPs are the major ones responsible for the metabolism of commonly prescribed drugs. Specifically, the most important isoforms are CYP1A1, CYP2C9, CYP2C19, CYP2D6, and CYP3A4. CYP3A4 has the widest substrate specificity and is estimated to be involved in the metabolism of ~50% of drugs used in humans.

Oxidations can occur at both aliphatic and aromatic carbon atoms. This leads to increased water solubility because of an increase in hydrogen bonding and ionizability. Oxidations can also occur at nitrogen and sulfur atoms with the formation of nitrogen and sulfur oxides. Another common phase I reaction is dealkylation; the removal of an alkyl group from either an oxygen producing an alcohol, or a nitrogen atom producing an amine. Reduction reactions are generally inhibited by oxygen and occur mainly under anaerobic conditions, e.g., in the gastrointestinal tract. Hydrolysis is catalyzed chiefly by esterases and peptides. Some of the most common phase I oxidation, reduction, and hydrolysis reactions are summarized in eqns [8]–[20].

Oxidations

Aromatic hydroxylation



For example, phenobarbital → *p*-hydroxyphenobarbital.

Aliphatic hydroxylation



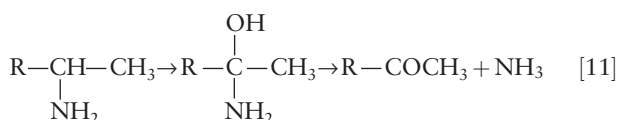
For example, pentobarbital → pentobarbital alcohol.

Dealkylation



For example, imipramine → desmethylinipramine (*N*-dealkylation) and codeine → morphine (*O*-dealkylation).

Oxidative deamination



For example, amphetamine → phenylacetone + NH₃.

N-oxidation



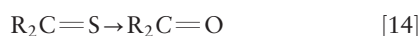
For example, chlorpromazine → chlorpromazine N-oxide.

S-oxidation



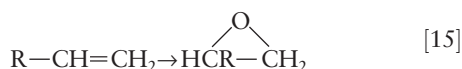
For example, chlorpromazine → chlorpromazine S-oxide.

Desulfuration



For example, thiopental → pentobarbital.

Epoxidation



For example, benzopyrene → benzopyrene epoxide.

Alcohol oxidation



For example, ethanol → ethanal.

Reductions

Azo reduction



For example, prontosil → sulfanilamide + triamino-benzol.

Nitro reduction



For example, *p*-nitrobenzoic acid → *p*-aminobenzoic acid.

Hydrolysis

Ester hydrolysis



For example, procaine → *p*-aminobenzoic acid + diethylaminoethanol and aspirin → salicylic acid + acetic acid.

Amide hydrolysis



For example, phenacetin → phenetidin + acetic acid.

Phase II metabolism Phase II conjugation reactions give rise to the bulk of the inactive, excreted products of a drug and involve the addition of an endogenous molecule to a compound. Generally, conjugation results in a large increase in water solubility and molecular weight, which encourages biliary excretion. Importantly, conjugation also leads to the inactivation of many reactive phase I products. The most common enzymes and functional groups involved are listed in Table 2.

Glucuronidation occurs at oxygen and nitrogen atoms and is one of the most common phase II reactions. It can occur with alcohols, phenols, hydroxylamines, carboxylic acids, amines, sulfonamides, and thiols and utilizes the energy-rich co-factor uridine diphosphoglucuronic acid (UDPGA).

Sulfation occurs at oxygen atoms. It can occur with phenols, alcohols, amines, and thiols and utilizes the energy-rich co-factor 3'-phosphoadenosine-5'-phosphosulfate (PAPS). Competition with glucuronidation for substrates can occur and in general sulfate conjugation predominates at low substrate concentrations and glucuronide conjugation at high concentrations, due to the kinetics of the two reactions and the more limited supply of PAPS compared to UDPGA.

Acetylation results in the addition of a CH₃CO group to nitrogen atoms to give an amide and although this results in no great change in water solubility the nitrogen is 'inactivated'. It utilizes the co-factor acetyl (CoA).

Glutathione conjugation is recognized as a protective compound useful for the removal of potentially toxic electrophilic compounds. It can occur

with epoxides, haloalkanes, nitroalkanes, alkanes, and aromatic halo- and nitro-compounds. The levels of glutathione in the body are finite and depletion following an overdose with a compound that undergoes this form of metabolism can result in toxicity due to the presence of large amounts of unconjugated reactive electrophiles, e.g., paracetamol.

Table 2 Principal conjugation reactions

Reaction	Enzyme	Functional group
Glucuronidation	UDP – glucuronyltransferase	–OH –COOH –NH ₂ –SH
Glycosidation	UDP – glycosyltransferase	–OH –COOH –SH
Sulfation	Sulfotransferase	–NH ₂ –SO ₂ NH ₂ –OH
Methylation	Methyltransferase	–OH –NH ₂
Acetylation	Acetyltransferase	–NH ₂ –SO ₂ NH ₂ –OH
Glutathione conjugation	Glutathione-s-transferase	Epoxide Organic halide

A good example of a compound undergoing a number of key conjugation reactions is *p*-aminosalicylic acid (**Figure 6**).

All tissues have some metabolic capacity, but greatest activity is seen in the liver, gastrointestinal lumen plus microflora, kidney, and blood. Many factors can affect drug metabolism including the dose level (exposure) and route, species and genetic differences, age, sex, diet, environment, disease, and the inhibition and induction of enzymes. An important exposure-related effect sometimes seen at high doses is when a metabolizing enzyme becomes saturated causing a change in drug:metabolite ratios and an altered pharmacological and/or toxicological response. The route of dose can also change the levels of total circulating drug, in particular, dosing by routes other than the oral route (e.g., intravenous, subcutaneous, inhaled) will result in first-pass metabolism being avoided.

Differences in the metabolic route between species are common, e.g., the rat mainly hydroxylates amphetamine leading to conjugated products whereas the rabbit and guinea pig (and man) mainly deaminate amphetamine. Acetylation of aromatic amines occurs in humans but not dogs while glucuronic conjugation is very poor in the cat. A good example of a compound that has the same routes of metabolism in different species but different rates is caffeine, as shown in **Table 3**. Species differences can be

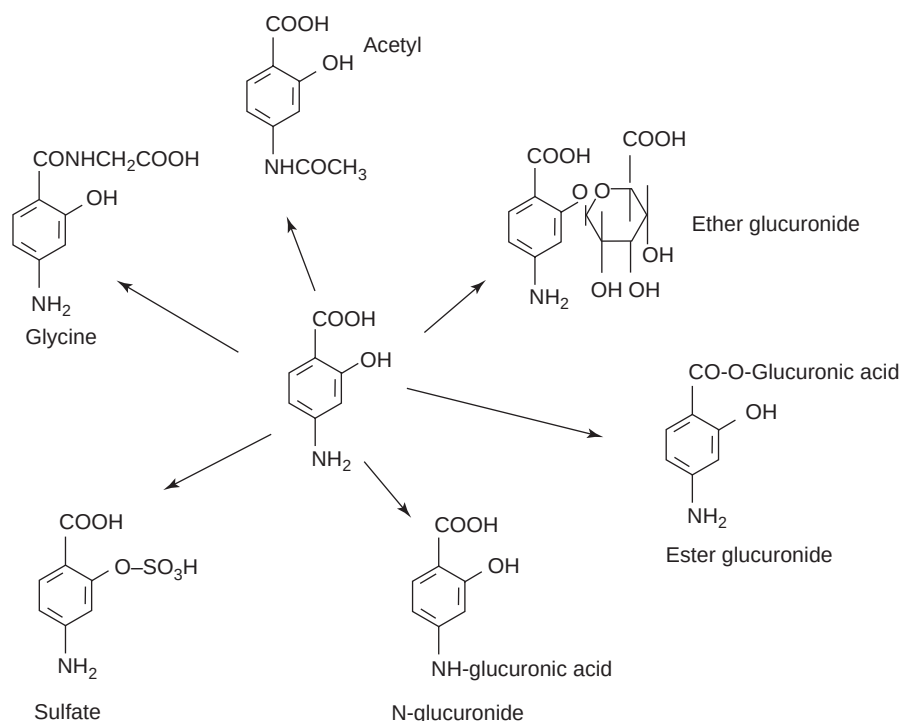


Figure 6 *p*-Aminosalicylic acid metabolism.

Table 3 Species differences in caffeine metabolism

Parameter	Man	Monkey	Rat	Rabbit
Total metabolism	322	235	160	137
Theobromine	28	13	15	19
Paraxanthine	193	11	20	42
Theophylline	16	190	20	30

Rates expressed as μ moles of product per min per mg protein. Reproduced with permission from Berthou *et al.* (1992) *Xenobiotica* 22: 671–680; © Taylor & Francis Ltd.; <http://www.tandf.co.uk/journals>.

important and usually animals are selected for toxicology studies on the basis of whether their metabolism most closely mimics the expected metabolism in humans.

Genetic differences within species (genetic polymorphism) may result in absences or deficiencies in enzymes, which may predispose individuals to adverse reactions. A genetic deficiency in the P-450 isoenzyme family (lack of CYP2D6) in a proportion of the population is why the antihypertensive debrisoquine is poorly metabolized in certain individuals (Figure 7).

A genetic polymorphism that leads to individuals having differences in their ability to acetylate compounds means that those classed as 'slow-acetylators' can have greater exposures to unconjugated reactive phase I products as has been observed for the anti-tuberculosis drug isoniazid.

The rates and routes of metabolism can vary with age and the lower rates and reduced routes of metabolism usually seen in the young and the old can make them more susceptible to drug action. Differences in hormonal levels, in particular, can lead to differences in metabolism between sexes. Dietary and environmental factors such as the presence of alcohol and cigarette smoke can affect metabolism, as can certain disease conditions, e.g., liver disorders, diabetes. Such conditions can lead to decreased enzyme activity, altered hepatic blood flow, and changes in plasma protein levels which, in turn, can affect circulating 'active' drug levels.

Drug–drug interactions Many compounds are able to 'inhibit' metabolizing enzymes, i.e., decrease their activity (cause an increase in steady-state plasma concentrations and a decrease in clearance). Inhibition can take place through the destruction or direct inhibition of the enzyme or the formation of a complex with it. Drug–drug interactions can occur when inhibition of the metabolism of a drug by a co-administered one leads to higher than expected levels of a pharmacologically active moiety. Also, the inhibition of some conjugation reactions can allow

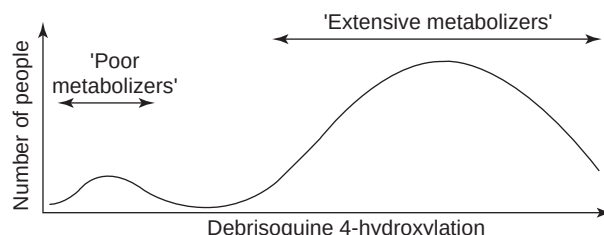
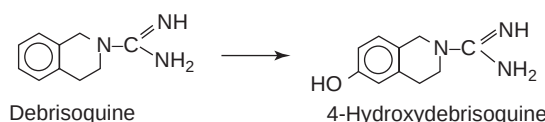


Figure 7 Distribution and rate of metabolism of debrisoquine in the Caucasian population. (Reproduced with permission from Gibson G and Skett P (1994) *Introduction to Drug Metabolism*, 2nd edn., ch. 4, p. 113; © Nelson Thornes Ltd.)

chemically reactive metabolites to escape detoxification and cause a toxic effect. A well-known example of a drug–drug interaction was the case of the non-sedating antihistamine terfenadine (a substrate for CYP3A4), which when co-administered with the antifungal ketoconazole, a potent inhibitor of CYP3A4, led to higher than expected plasma levels and cardiotoxic side effects. Another example is that of the stomach ulcer drug, cimetidine, which, when co-administered with the anticlotting drug warfarin, inhibited its rate of metabolism and dangerously lengthened blood clotting times.

Many compounds are able to 'induce' metabolizing enzymes, i.e., increase their activity (cause a decrease in steady-state plasma concentrations and an increase in clearance). The induction of metabolizing enzymes results in either an increase in their own metabolism (auto-induction) or that of other drugs (drug–drug interactions). It can reduce the toxicity of the parent molecule but lead to the enhanced formation of a chemically reactive metabolite and thereby overwhelm the detoxification processes. Sometimes, these metabolites can prove to be carcinogenic, e.g., polycyclic aromatic hydrocarbons. However, because enzymes are often induced in animals at higher doses than those used in man, the overall effect may not be clinically significant. On a purely practical level, enzyme induction may mean that it will be difficult to maintain efficacious levels and during regulatory toxicity studies it may not be possible to achieve high enough exposures to see toxicity. A number of CYPs are now usually monitored to assess the inductive capacity of test drugs including: CYP1A1, 2C, 2E, and 3A isozymes. For the reasons stated it is better to avoid developing drugs that are potent enzyme inducers, but, nevertheless, many important marketed

drugs are inducers, e.g., phenytoin, carbamazepine, phenobarbitone.

Assessment of ADME/PK

ADME/PK has a major role in the research and development of new drugs. The process of drug discovery can be divided into a number of distinct phases: hit screening and lead generation/optimization. The goal of the hit screening phase is to screen compound libraries to find molecules with a specific biological response, usually an *in vitro* potency. At the lead generation/optimization step the main aim is to apply structure activity relationships to improve upon these hits to produce leads that in turn are amenable to further optimization to produce the development candidate. The ADME/PK evaluation of compounds occurs throughout these phases but is particularly useful during lead generation/optimization. A schematic diagram of the discovery process from an ADME/PK perspective is shown in Figure 8.

Many methods are available for investigating key ADME/PK issues. Absorption issues include poor membrane permeability, extensive P-glycoprotein efflux, and poor physicochemical properties. Clearance/metabolism issues include extensive gut metabolism, extensive hepatic metabolism, enzyme induction and inhibition, and transporter-mediated excretion. Distribution issues include extensive protein binding and inadequate tissue penetration. *In silico* (computer-based), *in vitro* (cell-based), and *in vivo* (animal-based) techniques are available for investigating these issues. Structure-based *in silico* models and physicochemical methods allow basic predictions, while *in vitro* models offer a greater level of complexity and most closely mimic the *in vivo* situation. The most promising compounds from an *in vitro* potency and ADME/PK perspective are tested in appropriate *in vivo* pharmacology models. The aim of a typical drug discovery project is to identify a candidate molecule with suitable potency and selectivity against a target, sufficient oral bioavailability, metabolic stability, and duration of action, and minimal developability issues, e.g., potential for drug–drug interactions, ease of synthesis, dose linearity, etc.

Once a compound is selected for development, further studies are performed in animals and man to support the safety assessment and ensure that optimum dosing regimens are selected. Definitive *in vivo* PK and toxicokinetic studies are performed but also more detailed *in vitro* studies, particularly for the assessment of drug–drug interactions. ADME/PK support is continued throughout development (pre-clinical and phases I–IV) and a summary list of the main activities is shown in Table 4.

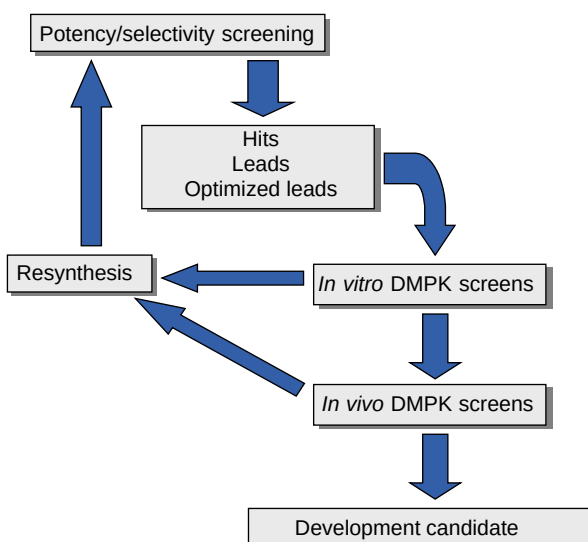


Figure 8 Schematic diagram illustrating the typical position of *in vitro* and *in vivo* DMPK screens in the lead generation/optimization phases of the drug discovery process. (Reproduced with permission from Roberts SA (2003): Drug metabolism and pharmacokinetics in drug discovery. *Current Opinion in Drug Discovery and Development* 6(1): 66–80; © Thomson Scientific.)

Table 4 Main ADME/PK activities performed during drug development

Toxicity testing – monitoring of TK in two nonhuman species
Dose ranging – tolerability over a range of doses
Dose linearity – check of PK proportionality between doses
Definitive PK – single and multiple dose PK
Sex differences – influence of gender on PK
Food interactions – influence of food on PK
Tissue distribution – animals only (usually radiolabeled drug)
Excretion balance – recovery in excreta (usually radiolabeled drug)
Metabolite identification – major routes of metabolism
PK/PD and dose response – determination of clear dose response relationships
Genetic polymorphisms – influence of genetic differences on metabolism
Effect of disease – PK in target population
Dose formulations – PK of final and new formulations
Drug interactions – effect of co-administered compounds on PK

In Silico Methods

Methods range from those based on molecular structure to those that use complex physiologically-based pharmacokinetic (PBPK) models requiring the input of experimentally determined data. Many physicochemical parameters can be generated automatically from chemical structure including H-bonding, molecular weight, log *P*, p*K_a*, polar surface area (PSA), and solubility and various software are available to do this. Lipinski's 'rule-of-five' is used by many companies to flag molecules likely to have poor absorption, i.e., compounds that have a molecular weight of

>500 , a calculated $\log P > 5$, number of H-bond donors >5 , and the number of H-bond acceptors >10 . PSA and rotatable bonds have also proven useful parameters for predicting absorption. A number of more complex models exist, e.g., for plasma protein binding, gastrointestinal absorption, and blood–brain barrier (BBB) penetration, that utilize several physicochemical parameters (calculated and/or experimentally determined). For instance, an algorithm for predicting brain penetration, $\log BB$ (the ratio of the steady-state concentrations of a compound in the brain and blood), designed using data on a series of brain penetrant compounds, requires PSA and calculated $\log P$ values as shown in eqn [21]:

$$\log BB = -0.0148\text{PSA} + 0.152C \log P + 0.139 \quad [21]$$

However, one drawback of attempts to model intestinal absorption and BBB penetration is that the effect of transporters is not generally accounted for as the factors that make a compound a substrate or inhibitor are not well understood at present.

Metabolism has proven a difficult event to predict *in silico* but some rule-based expert systems exist and some quantitative structure activity relationship models have been developed. The development of CYP models has been hindered by the lack of three-dimensional crystal structures; however, some homology models have been constructed for key CYP isozymes, e.g., CYP2C9, 2C19, 3A4, and 2D6. In conjunction with homology modeling, site-directed mutagenesis has been shown to help in understanding the relationship between substrate specificity and CYP structures. Important factors for substrate binding and enzyme selectivity include molecular size and shape, number and disposition of hydrogen bond donor/acceptor atoms, and number of aromatic rings in the molecule.

Physicochemical Methods

Measured physicochemical parameters are useful for the more accurate prediction of events such as drug absorption. $\log P$ is typically obtained by using potentiometric and spectrophotometric methods to measure the partitioning of a drug between an octanol and a water phase. Solubility is usually measured directly by monitoring achievable drug concentrations in test solutions. For discovery purposes, it is common to measure ‘kinetic’ solubility in which a dimethylsulfoxide solution of the compound is added to aqueous buffer and turbidimetry, nephelometry, and direct ultraviolet methods are used for analysis. However, drug dissolution profiles over time or as a function of pH are generally a better indicator of drug absorption than just inherent

solubility. In addition, a number of artificial permeability assays now exist that attempt to mimic the lipophilic nature of cell membranes and give an indication of absorption potential.

In Vitro Methods

The most commonly used cell culture model today is the 21-day Caco-2 human colonic cell line derived from a human adenocarcinoma. They can be cultured in special transwell cell culture plates (Figure 9) that enable the investigation of passive diffusion (apical to basolateral side) and active transport (basolateral to apical side). Although the system has its limitations, for many compounds it can give a good indication of likely *in vivo* absorption. An alternative cell culture, which has been shown to correlate well with absorption *in vivo* and permeability in Caco-2 cultures, is the 3-day Madin-Darby canine kidney cell line. Also, the expression of transporter proteins in cell cultures has led to new screens being established for identifying transporter substrates.

A variety of techniques are available for protein binding measurement including equilibrium dialysis, ultrafiltration, spectrofluorometric, and microdialysis methods.

Various cell culture models are available for assessing BBB penetration and although they can give some indication of potential brain penetration, they generally do not form sufficiently tight cell junctions.

A number of *in vitro* hepatic systems are available for assessing metabolism including liver slices, isolated hepatocytes, subcellular fractions (e.g., S9 and microsomes), and purified enzymes. Predictions of *in vivo* clearance can be made using *in vitro* intrinsic clearance (Cl_{int}) values derived from liver microsomes or hepatocytes. In addition to methods for assessing enzymic clearance, some exist for investigating drug efflux/uptake in the liver, e.g., isolated

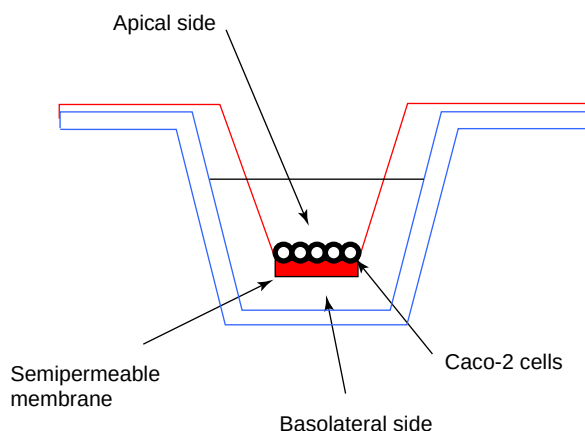


Figure 9 Schematic diagram of Caco-2 Transwell.

bile cannicular membranes and sandwich-cultured hepatocytes.

Mass spectrometry (MS) is now employed routinely for metabolite elucidation and in particular methods using quadrupole ion traps and time-of-flight liquid chromatography–MS (LC–MS) have proven very useful. Nuclear magnetic resonance (NMR) techniques are also being applied increasingly for more detailed structure elucidation. To aid in structure design, it is also useful to identify the P450 isoforms responsible for metabolism; cDNA-expressed enzymes and selective inhibitors can be used as well as isoform specific anti-P450 antibodies.

CYP inhibition assays include ones that utilize liver microsomes, isolated/cultured hepatocytes, and human cDNA-expressed enzymes. Typically, LC–MS/MS quantification is used for the microsome and hepatocyte methods while fluorimetric assays are used for the human cDNA-expressed enzymes. Typically, a weak inhibitor is defined as having a $k_i > 20 \mu\text{mol l}^{-1}$, whereas a potent inhibitor has a $k_i < 1 \mu\text{mol l}^{-1}$.

CYP induction assays are available that utilize liver slices, hepatocytes, and cell-based gene reporter constructs. *In vivo*, microsomes can be prepared from the livers of repeat-dosed test animals to examine if any key CYPs have been up-regulated. In discovery, the models utilizing hepatocyte cultures and cell-based gene reporter constructs have proven most useful. A number of analytical methodologies are available for supporting these various approaches including the LC–MS monitoring of probe compounds, mRNA analysis, and ribonuclease protection assays.

In addition to cell-based models, tissue-based models such as the Ussing chamber technique, the everted gut sac approach, and perfused isolated intestinal segments are also used, but only when it is important to understand the absorption processes in more detail. Unlike Caco-2, tissue-based models have the correct physiological levels of transporters and the presence of an apical mucus layer. Also, *in situ* and isolated organ perfusion methods exist for the gut, liver, lungs, kidneys, and brain and can provide data not directly obtainable *in vitro*. The isolated perfused liver is particularly useful since it allows an assessment of first-pass hepatic clearance, the quantitative distribution of metabolites in liver, blood, and bile, the effects of binding to plasma proteins and intracellular sites, and cellular uptake processes.

In Vivo Methods

Key PK parameters such as bioavailability, plasma elimination half-life, clearance, and volume of

distribution can be generated from PK profiles following oral and intravenous dosing. The efficiency of PK data generation has improved in recent years with the introduction of more sensitive higher throughput methods of analysis. The use of venous cannulated animals and multiple compound (cassette) dosing have contributed to the greater use of *in vivo* PK in drug discovery. Additional information can also be obtained in animals from biliary excretion studies through the cannulation of the common bile duct or gall bladder. Also, transgenic animals lacking or possessing particular gene sequences responsible for important drug metabolizing enzymes and transporters can be used to investigate specific issues. *In vivo* studies provide definitive PK data during development, aid in the design of *in vivo* pharmacology and toxicology studies, and help confirm (validate) the predictivity of *in vitro* methodologies and *in silico* models throughout the drug discovery process.

Methods available for measuring the concentrations of drugs in tissues and body fluids include radioimmunoassay, LC with ultraviolet detection (LC–UV), LC with radioactivity detection, and LC–MS. LC–MS has proven the most attractive method with its good sensitivity and specificity for a wide range of compounds as well as having a minimal sample handling requirement. In addition, MS and NMR techniques are used extensively for metabolite elucidation.

Techniques for quantitative ADME typically employ radiolabeled drugs to look at tissue distribution either in homogenized samples or by whole body autoradiography. Whole body autoradiography involving the location and quantification of radiolabeled material in thin frozen sections of tissue can be particularly useful. Plasma protein binding and metabolism studies also benefit from the use of radiolabeled material.

See also: **Drug Metabolism:** Metabolite Isolation and Identification; Isotope Studies. **Pharmacokinetics:** Absorption, Distribution, and Elimination; Pharmacodynamics.

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Metabolite Isolation and Identification

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Introduction

The study of the metabolism of new chemical entities is very important in selecting drug candidates, and increasingly metabolite isolation and identification studies are performed in the early phase of drug discovery. Drug metabolism studies provide vital information to facilitate the understanding of the pharmacological and toxicological action of drugs and satisfy regulatory requirements. The most common types of drug metabolism assays involve isolation and identification of metabolites generated *in vitro* and *in vivo*. The advent of several new technologies has facilitated the isolation and identification of metabolites, which are often formed in very small quantities in drug metabolism models. For example, the increased use of automation in bioanalysis and the availability of robust mass spectrometers have increased the rate at which data can be generated for elucidation of metabolic pathways. In this article, the current approaches and emerging technologies used for metabolite isolation and identification from various biological matrices are discussed.

The Importance of Drug Metabolism Studies

The fate of a drug after administration in animals and humans involves absorption, distribution, metabolism, and excretion (ADME). The body recognizes a

drug as a foreign compound and in most cases the drug is converted to a metabolite through multiple enzymatic pathways. Drug metabolism controls the levels of drugs in the body and may detoxify or activate a drug. In a nutshell, the goals of metabolite isolation and identification include (1) determination of routes of elimination, (2) estimation of rates of metabolic reactions, (3) assessment of metabolic stability, (4) testing of metabolizing enzyme inhibition, (5) structural characterization of formed metabolites, and (6) evaluation of the biological activity of metabolites. Morphine, for example, is converted to morphine-6-glucuronide, which is partially responsible for its pharmacological activity in humans. The various biological outcomes of drug metabolism are shown in **Figure 1**. In recent years, regulatory authorities have also demanded information on metabolite identity and biological activity before new drug applications are approved. In fact, the successful drug candidates must have acceptable ADME, pharmacological, and toxicological profiles. Thus, the role of metabolite isolation and identification studies in selecting better lead compounds with acceptable ADME properties *in vivo* is pivotal in drug discovery and regulatory filing for new drugs.

Models for Drug Metabolism Studies

Both *in vitro* and *in vivo* models are used to optimize compounds for favorable metabolic properties during drug discovery and development. After administration of a drug to animals and humans, the drug may be transferred from the blood into the liver, the major site of drug metabolism in the body, where it is metabolized or excreted into the biliary system.